

电子自旋

(Spin of electron)

一、斯特恩 —— 盖拉赫实验 (1921)

The Stern-Gerlach Experiment

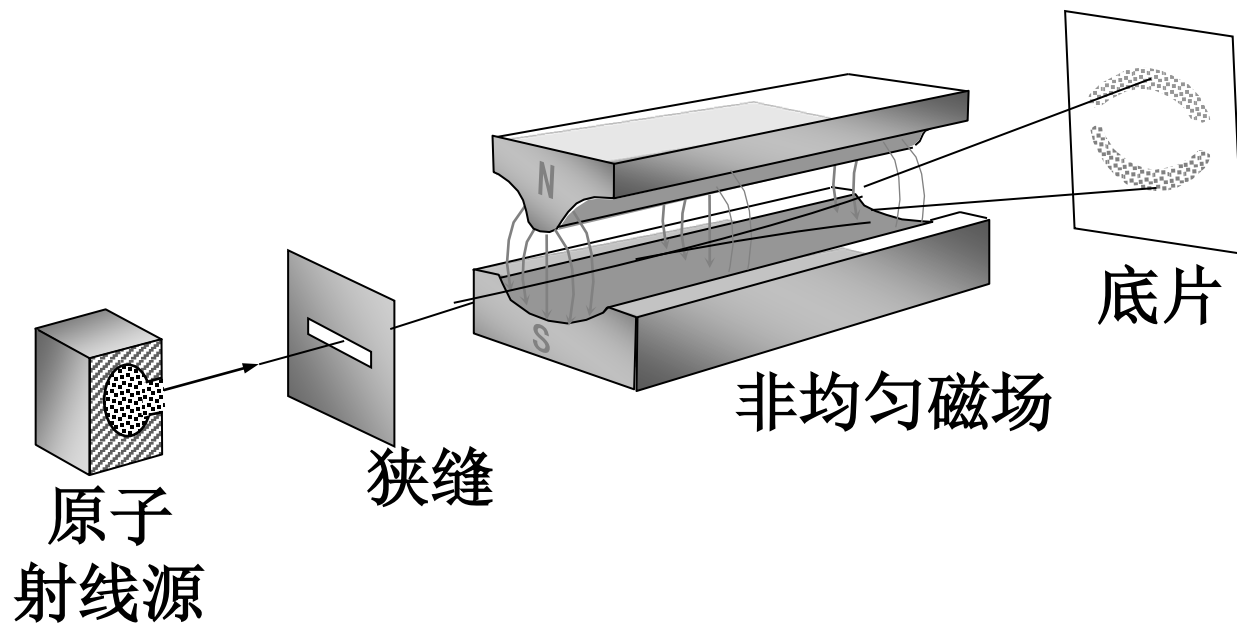


Otto Stern
(1888-1969)



Walter Gerlach
(1889-1979)

实验装置:



由于电子带电，电子绕原子核作轨道运动，就相当于一个闭合载流线圈一样。

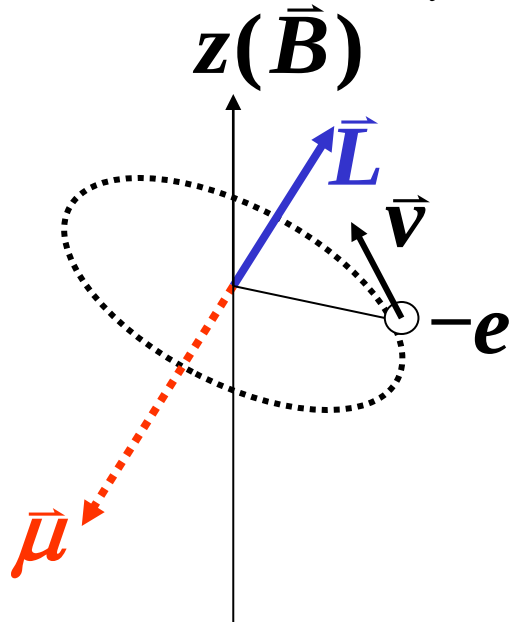
其磁矩大小为 $\mu = iS$ 电流强度

$$i = \frac{v}{2\pi r} e$$

载流线圈面积 $S = \pi r^2$

所以有

$$\mu = \frac{ve}{2\pi r} \pi r^2 = \frac{e}{2m} mvr = \frac{e}{2m} L$$



由于电子带负电，角动量与磁矩的矢量关系为

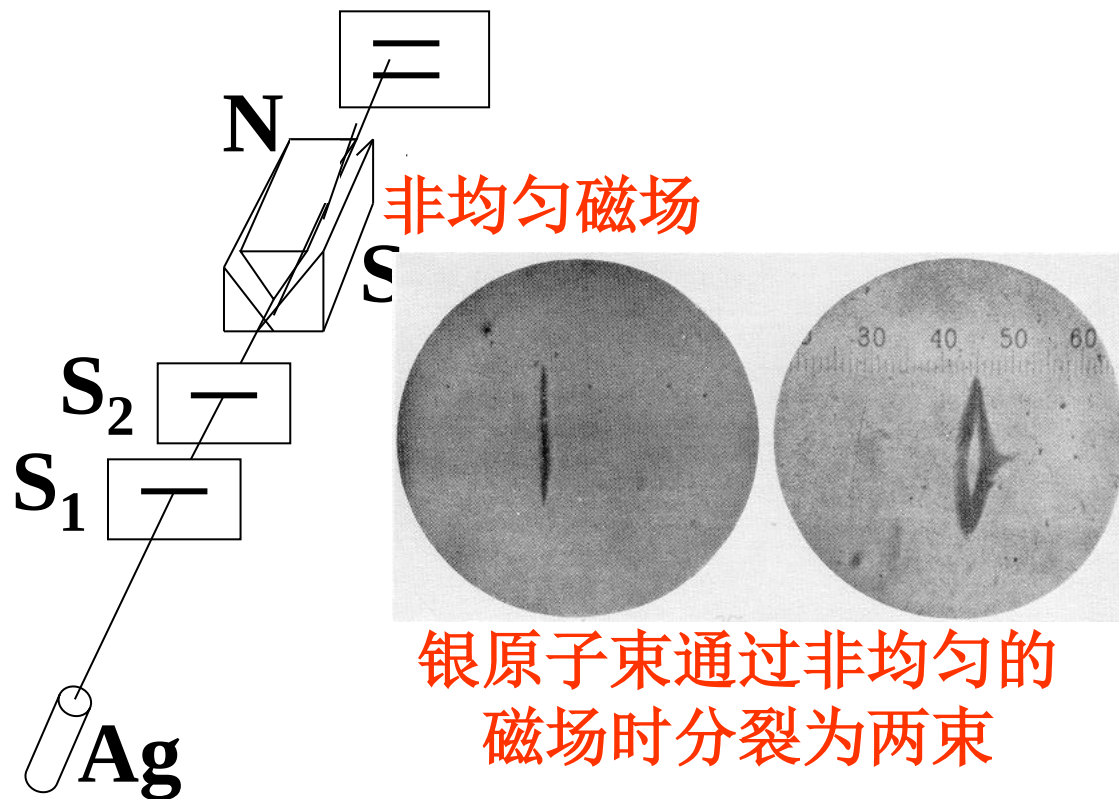
$$\vec{\mu} = -\frac{e}{2m} \vec{L}$$

由于角动量是空间量子化的，那么磁矩也是空间量子化的，它们在空间 z 方向的投影也是量子化的，应有

$$\mu_z = -\frac{e}{2m} L_z = -m_l \frac{e\hbar}{2m}$$

实验结果： 无法解释！

基态 ($l = 0$) Ag 原子的沉积是 2 条，不是奇数 $2l + 1 = 1$ 条。



斯特恩正在观测



二、电子自旋

1925年，两位当年的荷兰学生乌伦贝克和哥德斯密特在分析上述实验的基础上提出了大胆的看法：

- (1) 电子不是一个质点，它存在一种内秉的运动 —— 自旋，相应地有自旋角动量和自旋磁矩。
- (2) 电子自旋角动量 $\vec{S}$ 的大小类似于“轨道”角动量为
 $|\vec{S}| = S = \sqrt{s(s+1)}\hbar$ —— s 称为自旋量子数
- (3) 电子自旋角动量在空间相对外磁场方向的取向也是空间量子化的。

电子轨道角动量 $\vec{L}$

大小： $L = \sqrt{l(l+1)}\hbar$

其中 $l = 0, 1, 2, \dots (n-1)$

在外磁场中 $\vec{L}$ 有 $2l+1$ 个取向。



电子自旋角动量 $\vec{S}$

大小： $S = \sqrt{s(s+1)}\hbar$

其中 $s = ?$

在外磁场中只有二个取向。

则 $2s+1=2$

$$\therefore s = \frac{1}{2}$$

所有原子的自旋角动量都相同，
故不再当做一个量子数提出。

$$S = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)}\hbar = \sqrt{\frac{3}{4}}\hbar$$

对比

轨道角动量 $\vec{L}$ 在外
磁场中投影: $L_z = m_l \hbar$

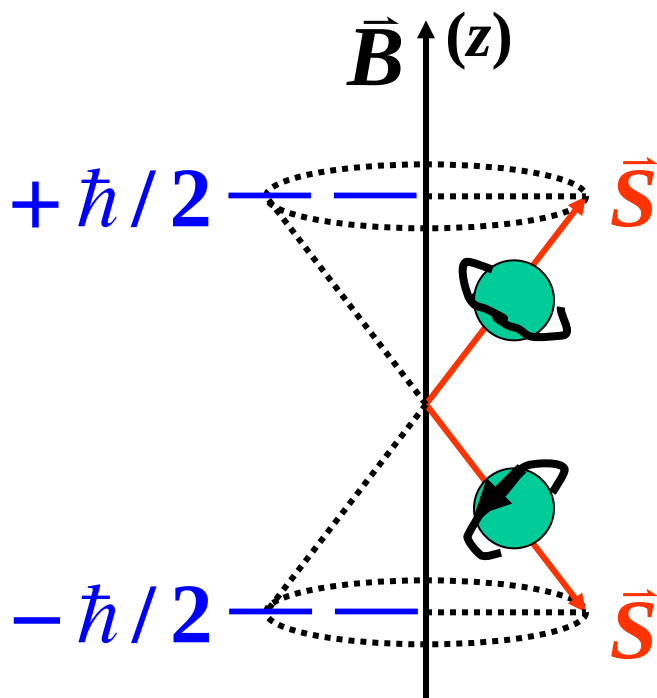
$$m_l = 0, \pm 1, \pm 2, \dots \pm l$$

自旋角动量 $\vec{S}$ 在外
磁场中投影: $S_z = m_s \hbar$

$$m_s = \pm s = \pm \frac{1}{2}$$

m_s —— 为自旋磁量子数。

$$\therefore S_z = \pm \frac{1}{2} \hbar$$



当轨道动量矩 $l = 0$ ，但有自旋磁矩。
自旋磁矩只有二个指向，因此银原子
射线分裂为二条。

(泡利、洛伦兹等的反对) “You are both young enough to
(埃伦菲斯特的支持) allow yourselves some foolishness!”

电子自旋角动量 $\bar{S}$ 的大小为

$$S = \sqrt{s(s+1)}\hbar$$

$$s = \frac{1}{2} \quad \text{称为自旋量子数}$$

$$S = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)}\hbar = \sqrt{\frac{3}{4}}\hbar$$

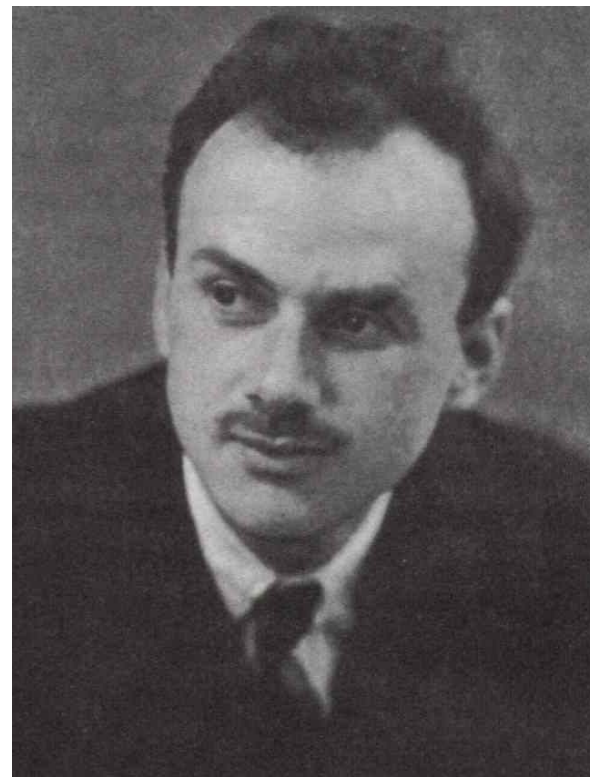
自旋角动量 $\bar{S}$ 在外磁场中

投影: $S_z = m_s \hbar$

$$m_s = \pm s = \pm \frac{1}{2} \quad \text{为自旋磁量子数}$$

$$\therefore S_z = \pm \frac{1}{2} \hbar$$

当轨道角动量 $L = 0$ ，但有自旋磁矩。
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P. Dirac
(1902–1984)

The Story of Spin

"it is indeed very clever but of course has nothing to do with reality".



Wolfgang Pauli
(1900-1958)



Ralph de Laer Kronig
(1904-1995)

**First suggested by Kronig in early 1925;
greeted with great hostility by Pauli.
Idea was never pursued further by Kronig.**

The Story of Spin

**George Uhlenbeck
(1900-1988)**



**Samuel Goudsmit
(1902-1978)**

A few months later, two young Dutch research students of Ehrenfest in Leiden — Uhlenbeck and Goudsmit, had the same idea. They produced a short paper in 1925 and gave it to Ehrenfest.

The Story of Spin

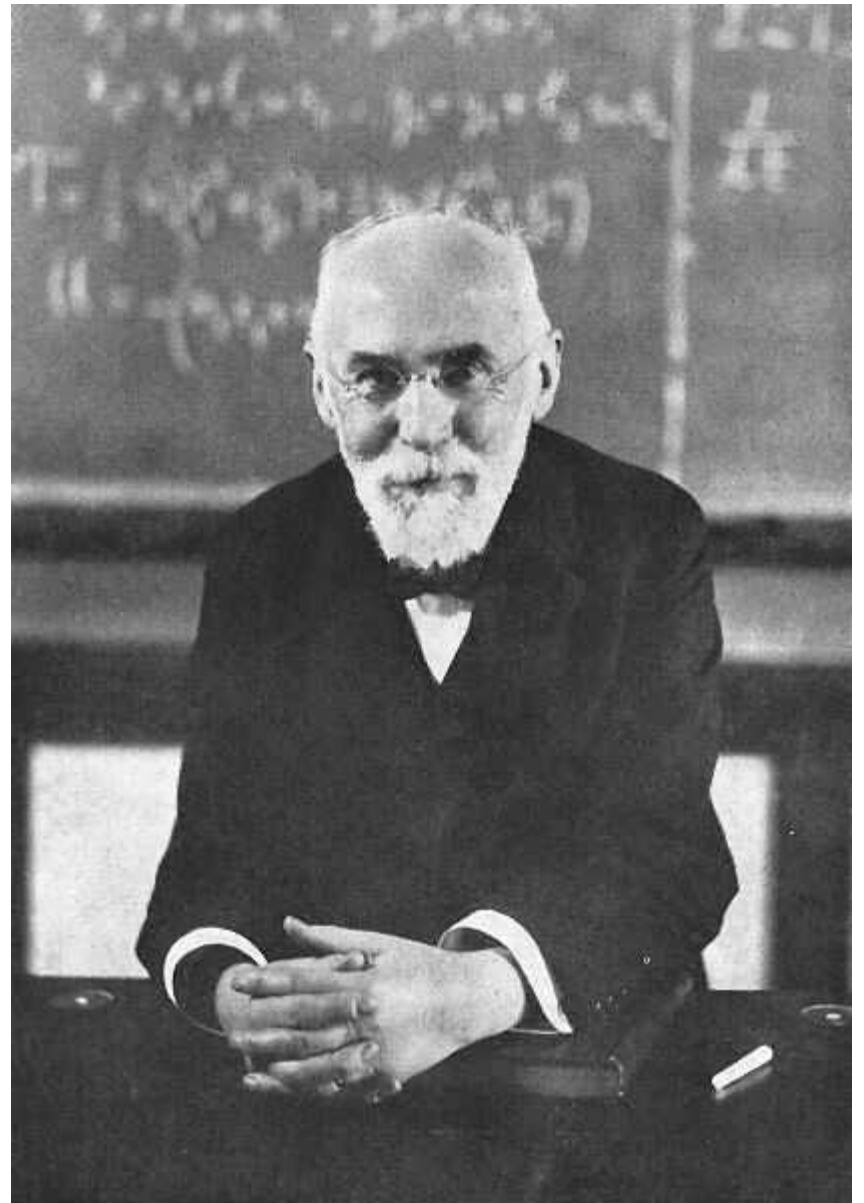
**Paul Ehrenfest
(1880-1933)**



Ehrenfest liked the idea very much. He suggested to Uhlenbeck and Goudsmit that they should go and talk it over with Hendrik Lorentz, who lived close by in Haarlem. They did go and talk it over with Lorentz.

The Story of Spin

Lorentz said, "No, it's quite impossible for the electron to have a spin. I have thought of that myself, and if the electron did have a spin, the speed of the surface of the electron would be greater than the velocity of light. So, it's quite impossible."



**Hendrik Lorentz
(1853-1928)**

The Story of Spin

Uhlenbeck and Goudsmit rushed back to Ehrenfest and said they would like to withdraw the paper that they had given to him. But Ehrenfest said, "No, it's too late; I have already sent it in for publication.", and he added: "**You are both young enough to allow yourselves some foolishness!**"

**Wolfgang Pauli and
Paul Ehrenfest**



The role of chance and status in the development of physics knowledge — the discovery of the electron spin

I think that you [Goudsmit] and Uhlenbeck have been very lucky to get your spinning electron published and talked about before Pauli heard of it. It appears that more than a year ago Kronig believed in the spinning electron and worked out something; the first person he showed it to was Pauli. Pauli ridiculed the whole thing so much that the first person became also the last and no one else heard anything of it.

Part of a letter by L. H. Thomas to Goudsmit (25 March 1926)

小结

电子自旋磁量子数 $m_s = \pm \frac{1}{2}$

决定电子自旋角动量在外磁场中的两种指向， $S_z = m_s \hbar$ 也影响原子在外磁场中的能量。

电子自旋角动量的大小

$$S = \sqrt{\frac{3}{4}} \hbar$$

斯特恩-盖拉赫实验：只能用电子自旋角动量的空间取向量子化来解释，是直接证实电子自旋存在的最早实验之一。